

Ex. No. 3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss

AIM

To analyze the propagation loss characteristics of **Sub-1 GHz, Mid-band, and mmWave** frequency bands using the Free Space Path Loss (FSPL) model in MATLAB and compare their performance under different communication conditions.

Objective 1: Analyze Propagation Loss with Distance

Aim

To compute and compare the propagation loss of **Sub-1 GHz (900 MHz), Mid-band (3.5 GHz), and mmWave (28 GHz)** over varying communication distances using the Free Space Path Loss (FSPL) model.

MATLAB Code

```
clc; clear; close all;
```

```
d = 0.1:0.1:5; % Distance (km)
```

```
f = [900 3500 28000]; % Frequency (MHz)
```

```
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
```

```
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
```

```
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(d,PL1,'b','LineWidth',2); hold on
```

```
plot(d,PL2,'r','LineWidth',2)
```

```
plot(d,PL3,'k','LineWidth',2)
```

```
title('Propagation Loss vs Distance')
```

```
xlabel('Distance (km)')
```

```
ylabel('Loss (dB)')
```

grid on

% ----- Graph 2 (NO legend used to avoid error) -----

subplot(2,1,2)

Freq = [900 3500 28000];

Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq, Loss, '-o', 'LineWidth', 2)

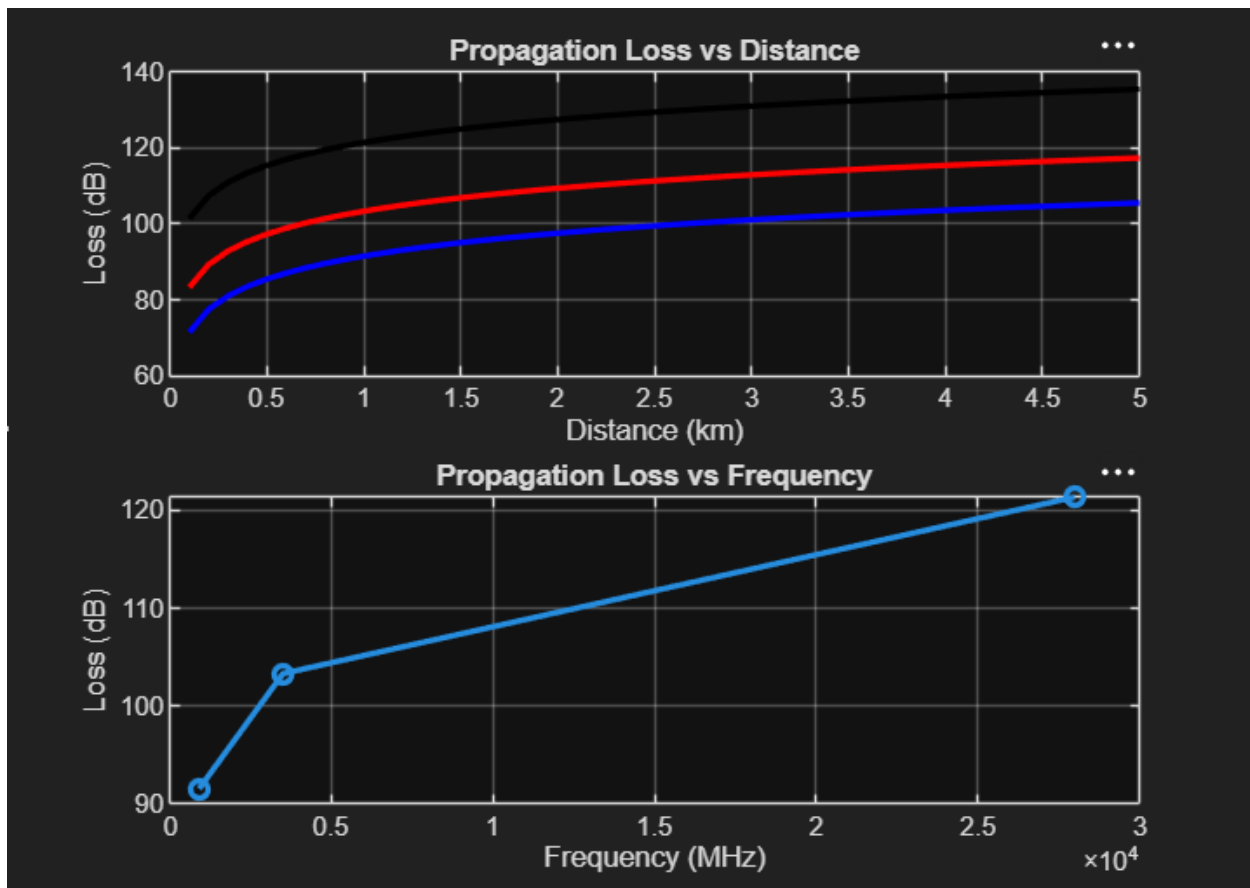
title('Propagation Loss vs Frequency')

xlabel('Frequency (MHz)')

ylabel('Loss (dB)')

grid on

Output



- Propagation Loss vs Distance
- Propagation Loss vs Frequency

Result

The propagation loss increases as the communication distance increases. Among the three frequency bands, **mmWave (28 GHz)** experiences the highest propagation loss, while **Sub-1 GHz (900 MHz)** experiences the lowest loss, making it more suitable for long-range wireless communication.

Objective 2: Analyze the Effect of Operating Frequency

Aim

To evaluate the effect of different operating frequencies on propagation loss at a fixed communication distance using the Free Space Path Loss model.

MATLAB Code

```
clc; clear; close all;

d = 2; % fixed distance (km)
f = [900 3500 28000]; % MHz
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure
% ----- Graph 1: Frequency vs Loss -----
subplot(2,1,1)
plot(f,PL,'-o','LineWidth',2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
% ----- Graph 2: Band Comparison -----
subplot(2,1,2)
bar(PL)
title('Comparison of Propagation Loss Across Bands')
```

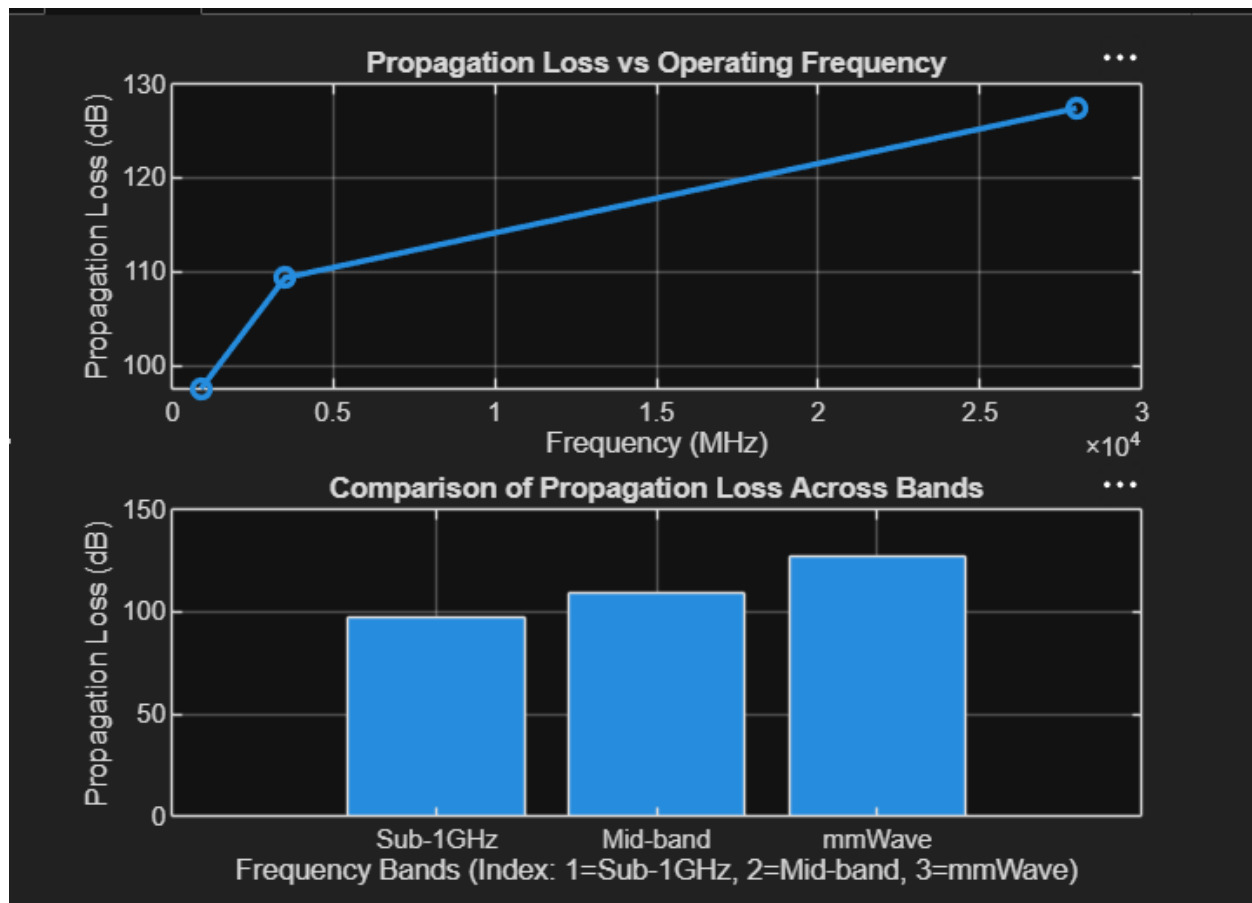
```
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
```

```
grid on
```

Output



- Propagation Loss vs Operating Frequency
- Comparison of Propagation Loss Across Bands

Result

The propagation loss increases with increasing operating frequency. The **mmWave band** shows the highest attenuation, whereas the **Sub-1 GHz band** provides better signal coverage and lower propagation loss.

Objective 3: Compare Propagation Loss Across Spectrum Bands

Aim

To compare the propagation loss of **Sub-1 GHz, Mid-band, and mmWave** spectrum bands at a fixed communication distance and analyze their suitability for wireless communication.

MATLAB Code

```
clc; clear; close all;

% Frequency bands (MHz)
f = [900 3500 28000];

% Distance (km)
d = 2;

% Propagation Loss (FSPL)
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

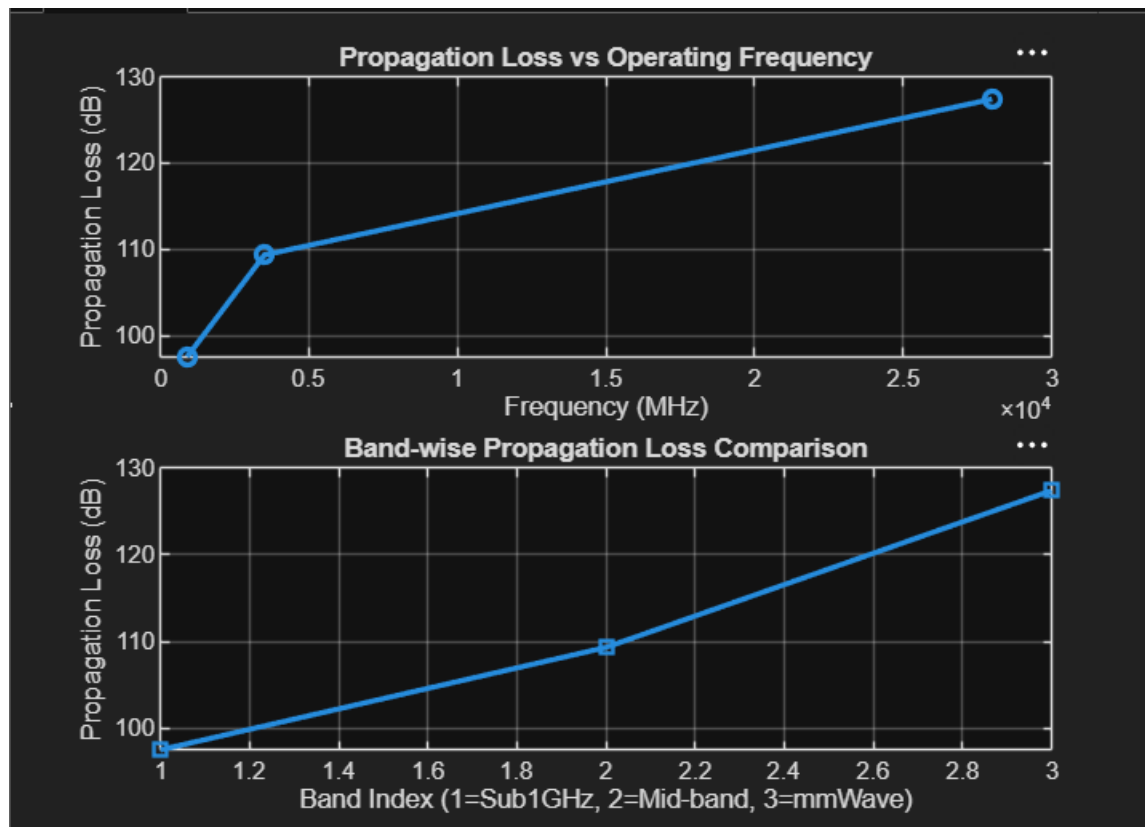
% ----- Graph 1: Simple Line Plot (SAFE) -----

subplot(2,1,1)
plot(f, PL, '-o', 'LineWidth', 2)
title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on

% ----- Graph 2: Safe Comparison Plot -----

subplot(2,1,2)
plot(1:3, PL, '-s', 'LineWidth', 2)
title('Band-wise Propagation Loss Comparison')
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
grid on
```

Output



- Propagation Loss vs Operating Frequency
- Band-wise Propagation Loss Comparison

Result

The comparison shows that propagation loss is lowest for the **Sub-1 GHz** band, moderate for the **Mid-band**, and highest for the **mmWave** band. Hence, Sub-1 GHz is more suitable for wide-area coverage, Mid-band offers a balance between coverage and capacity, and mmWave is best suited for high-speed, short-range communication due to its higher attenuation.